



Research review paper

Renaissance of protein crystallization and precipitation in biopharmaceuticals purification



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ABSTRACT

The current chromatographic approaches used in protein purification are not keeping pace with the increasing biopharmaceutical market demand. With the upstream improvements, the bottleneck shifted towards the downstream process. New approaches rely in Anything But Chromatography methodologies and revisiting former techniques with a bioprocess perspective. Protein crystallization and precipitation methods are already implemented in the downstream process of diverse therapeutic biological macromolecules, overcoming the current chromatographic bottlenecks. Promising work is being developed in order to implement crystallization and precipitation in the purification pipeline of high value therapeutic molecules. This review focuses in the role of these two methodologies in current industrial purification processes, and highlights their potential implementation in the purification pipeline of high value therapeutic molecules, overcoming chromatographic holdups.

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1. The new trend: anything but chromatography

Recent developments in more efficient fermentation and cell culture technologies, led to an improvement in biopharmaceutical product titers from an average of 0.2 g/L in 1985 to 2.56 g/L in 2014, although monoclonal antibodies (mAbs) titers can reach 27 g/L (Perciva, 2008). This improvement in the upstream of biopharmaceuticals production

led to a shift in the manufacturing bottleneck towards the downstream processing (DSP), which can weigh up to 70% of the total production costs (Broly et al., 2010). Currently, DSP face several challenges, namely the need to (i) deal with the high upstream titers and production volumes, (ii) accomplish high purities and yields with a reduced number of unit operations through process integration and intensification and (iii) comply with the trend of single-use technology DSP (Kuczewski et al., 2011; Shukla and Gottschalk, 2013).

For biopharmaceutical products with high added value, DSP design aims to achieve high quality, concentration, productivity and yield. DSP trains of biological products usually comprise four main steps:

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recovery, isolation, capture and polishing (Broly et al., 2010). The capture and polishing steps are frequently based on chromatographic operations with high costs associated with resin price (e.g. 5,000–14,000 €/L (Franzreb et al., 2014)) and operation costs (e.g. high buffer consumption of 300,000–2,000,000 L/batch). The development of new and more efficient upstream processes, leading to higher protein titers, constrains the chromatographic process of the capturing step (Gottschalk, 2006). As a result, new approaches to DSP design are being considered, due to the high costs associated. High-throughput (HTP) screenings are used in combination with statistically planned experiments (Design of Experiments, DoE) to recognize the influence of different factors, thus narrowing down the experimental conditions to be tested. In addition, Quality by Design (QbD) can also be implemented with HTP or DoE (Pathak et al., 2014; Zhang et al., 2014), combining in this way the manufacturing process with product quality, while improving process robustness and productivity.

Considering mAbs DSP, the conserved domains of mAbs led many companies to define a general purification process based on a common sequence of unit operations. The heart of the process is the protein A chromatography capture step usually followed by two polishing steps composed by ion exchange chromatography. Although stable, reliable, and reproducible, protein A chromatography is considered to be a productivity bottleneck, requiring several cycles per batch (e.g. 1–8 cycles/batch mAb (Franzreb et al., 2014)), and being particularly expensive. In order to overcome the limitations demonstrated by chromatography such as high resin cost, diffusion and capacity limitations, new approaches based on Anything but Chromatography (ABC) methodologies are gaining increasing interest (Barroso et al., 2014; Dhadge et al., 2014b). Currently, new trends in non-chromatographic methodologies for DSP include membrane processes (Dong and Bruening, 2015), liquid–liquid phase extraction (Jacinto et al., 2015), magnetic separation (Batalha et al., 2010; Dhadge et al., 2014a; dos Santos et al., 2016), precipitation and crystallization (Gronemeyer et al., 2014). Non-chromatographic methods aim to reduce or even replace chromatographic operations in the purification pipeline, which can achieve high yields and purity, despite its disadvantages. On the other hand, non-chromatographic methods can cope with the high titers and volume feed produced upstream. Within the pool of non-chromatographic methods available, precipitation is being applied to the purification of some biopharmaceuticals (Hammerschmidt et al., 2015a; Sim et al., 2012a; Sommer et al., 2014), and protein crystallization is a new and challenging method that can overcome the current chromatographic challenges (Smejkal et al., 2013; Zang et al., 2011). Although crystallization and precipitation are currently being used for the purification of commercially available proteins, such as ovalbumin (Judge et al., 1995), lipase (Jacobsen et al., 1998), elastase (Lewis et al., 1956), protease I and II (Srivastava and Aronson, 1981) and lactose (Commins et al., 1980) as revised by Miranda et al., 2009, these are still not widely implemented in the industrial biopharmaceutical sector despite their potential (Gagnon, 2012).

2. Understanding protein precipitation and crystallization

Protein precipitation is based on the premise that a homogenous protein solution is only homogeneous up to a certain protein concentration from which a new phase, corresponding to the precipitate formation, starts to appear. By changing the environmental conditions, one expects to exceed the protein solubility limit, leading to the formation of the new solid phase. Protein crystallization is a form of precipitation in which the precipitate is formed in an ordered manner.

Protein precipitation is mostly used for low value products. However, due to its advantages in DSP, research has been conducted towards high value products, such as therapeutic proteins in the biopharmaceutical industry.

Protein precipitation depends on the solubility of a protein in an aqueous media, which is deeply related with the distribution of the

hydrophobic and hydrophilic amino acid residues at the protein surface (Englund and Seifter, 1990). In a time where proteins were considered as colloids, the first protein precipitation experiments were carried by Antoine Fourcroy in circa 1789. Fourcroy was able to precipitate gelatin by boiling various animal by-products and albumin from egg white, blood serum and milk casein by heating, acid or alkaline precipitation (Tan and Yip, 2009). The following studies focused on proteins that could be obtained in large quantities, namely from blood and egg white. The first studies using precipitation as a means for protein purification were developed by Edwin Joseph Cohn during World War II (1939–1945). He became famous for the separation of blood components by fractional precipitation. The Cohn fractionation process aimed at obtaining albumin to be administered to soldiers with blood loss, starting from blood plasma and taking advantage of the different solubility among the different plasma proteins (Cohn, 1941). This also showed that a protein purified by precipitation had biological activity. Since its discovery, protein precipitation has been used in proteins DSP, for purification and concentration purposes (Hinderer and Arnold, 2012; Sheth et al., 2014a).

With the development of single-crystal X-ray crystallography, the crystallization of macromolecules became one of the most common methods to determine protein structures and understand fundamental biochemical pathways or even design new drugs. However, by the time the first protein was crystallized, in the first half of the 19th century, crystallization was applied as a purification and biophysical characterization method (Giegé, 2013; Hünefeld, 1840). Nonetheless, the interest in crystallization from a purification point of view increased due to the low costs associated, the ability to cope with high volumes, and with high concentrations of the target molecule. In addition, crystallization in the DSP of biopharmaceuticals has the ability to integrate protein purification, stabilization and concentration in one single step.

Macromolecular crystallization can be performed by four different methods, namely vapor diffusion, microbatch, dialysis and free interface diffusion. In the vapor diffusion the protein sample is set to equilibrate with a reservoir solution that contains similar precipitant and buffer solution at a higher concentration. In batch crystallization, the supersaturated protein sample is mixed with the precipitants in the same mixture and then placed in a mineral oil of branched alkanes in the C20+ range allowing little to none water diffusion, in this way enabling a controlled evaporation. Crystallization by dialysis, makes use of dialysis membranes where ions can pass but not proteins or polymers, in this way slowly supersaturating the protein solution leading to crystal formation (Thomas et al., 1989). Free interface diffusion relies on carefully placing the precipitant solution on top of the protein solution in a capillary. Due to the narrow diameter of the capillary, the precipitant and protein will slowly inter-diffuse until equilibrium is reached, leading to protein supersaturation (Salemme, 1972).

Although a rational approach might be drawn the fact is that both precipitation and crystallization experiments have a rather empirical design, mainly due to the complexity of the occurring phenomena (Altan et al., 2016). Currently, there is no comprehensive theory behind the process in order to guide the experimental design. Furthermore, the wide range of macromolecules possible to crystallize enhances the complexity of the crystallization design. The guidelines for macromolecular crystallization involve a systematic search, changing individual parameters that can influence the crystallization process. Once found the set or multiple sets of parameters that yield the best precipitate/crystal outcome, an optimization can be performed to obtain the best possible crystals, which for crystallography purposes correspond to single, edge-shaped crystals.

Nonetheless, some theories have been proposed to understand protein behavior in an aqueous solution. Classical nucleation theory (CNT) aims to calculate the nucleation rate of a particular system, as a theoretical model this is an approximation (Lutsko and Durán-Olivencia, 2013; Sleutel et al., 2014). The phase diagram plots the protein concentration versus any adjustable parameter (e.g. temperature, precipitant

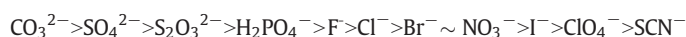
concentration, salt concentration) (Fig. 1A). These diagrams enlighten the kind of interactions occurring between proteins in a qualitative manner, but also in a quantitative way it is possible to recover the enthalpy and entropy values for the protein in the aqueous solution (Asherie, 2004; Vekilov, 2010).

When designing a crystallization assay it is necessary to know (as well as possible) the target protein nature and the interactions which most probably enable crystal formation (Hekmat, 2015). Protein nature, i.e. structure and amino acid composition, have a great influence in the outcome, e.g. intrinsically disordered proteins tend to have more polar or highly charged groups than hydrophobic residues. In addition, a biochemical characterization of the target protein also helps designing a target precipitation protocol (Gavira, 2015). Moreover, the protein purity and the purification process may also present some challenges if the main goal is X-ray diffraction for structure determination (Gavira, 2015).

The variables that have to be considered when designing a crystallization strategy can be divided into physical, chemical and biochemical, and range from the precipitant to the environmental vibrations (McPherson and Gavira, 2014). In general, protein concentration, temperature or temperature variation, precipitant concentration, the presence/absence of ligands and the salt/buffer used play a greater role than gravity, viscosity or electric/magnetic field.

Ion-ion, ion-dipole, dipole-dipole and hydrophobic interactions contribute all together to protein stability (Englund and Seifert, 1990). Dipolar substances, as water, will compete with dipolar interactions; ions may disrupt ionic interactions, hydrophobic substances will seek for the hydrophobic core. The sum of all medium modifications, when maintained constant, will lead to a protein rearrangement into a stable conformation. Variations of temperature, pH, salt concentration, presence of organic solvents and polymers can trigger precipitation processes. For example, a rise in temperature favors the formation of hydrophobic interactions at the same time as it weakens dipolar interactions; a decrease in temperature leads to the opposite effect. Meanwhile when a protein surface does not have net charge – at its isoelectric point – it presents the least surface hydration, with the minimum solubility while in the nondenatured form. It should be noted that minimum solubility is not synonymous of insolubility/precipitation, as precipitation requires additional perturbations. The addition of salt also plays an important role in protein precipitation/crystallization. Along the years, protein precipitation/crystallization appears to follow the Hofmeister series (Collins, 2004). This series ranks the ions according to their relative effect on protein solubility and structure stability in an aqueous system (Zhang and Cremer, 2006). In general, the effect of

anions is more pronounced than that of cations. The Hofmeister anion series is as follows (Zhang and Cremer, 2010, 2006):



To the left of Cl^- , are located the kosmotropic anions, on the other side to the right are the chaotropic anions (Collins, 2004; Zhang and Cremer, 2006). Kosmotrope species are referred to as ‘water structure makers’ for strengthening bulk water hydrogen bonding network, these species also stabilize and have a salting-out effect on proteins. On the other hand, chaotropic species are cited as ‘water structure breakers’ by weakening the bulk water hydrogen bonding network, unlike kosmotropes, chaotropes destabilize and tend to unfold proteins leading to a salting-in effect (Collins, 2004; Zhang and Cremer, 2010, 2006). The concentration of salt used will vary according to the protein under study and its surface net charge, size and shape. Organic solvents such as methanol, butanol, ethanol and acetone, are commonly used in precipitation studies, all containing both hydrophilic and polar domains. While the polar groups compete with water to interact with the protein polar groups, the hydrophobic domains disrupt intramolecular interactions, resulting in a solubility decrease triggering protein precipitation. Once again, the concentration of the organic solvent will vary according to the protein under study, as for salt precipitation. Nonetheless, the use of organic solvents for protein precipitation might lead to protein denaturation; in alternative, different polymers can be used. Polymers are frequently used in protein precipitation, the most commonly used being polyethylene glycol (PEG), usually 20% (w/v) of PEG with molecular weights >4000 Da due to the higher efficacy. In solution, PEG not only attracts water molecules from the protein surface (McPherson, 1976), but also has an effect in sterically exclude proteins from its hydrodynamic volume (Atha and Ingham, 1981; Bolen, 2004), enhancing protein-protein interactions leading to protein precipitation. Although limited by the availability and stability of the target protein, different combination of precipitants can be used in an initial screening (Kowacz et al., 2015; Ribeiro et al., 2014). Nowadays, different screenings have been developed and are commercially available (e.g. JBScreen Wizard from Jena Bioscience) and some can be developed in-house, usually by macromolecular crystallography research labs.

In biologics DSP, precipitation is presented as an alternative to the current chromatographic methods. Precipitation protocols have a low cost associated, and can be performed with high protein titers and volume. Moreover, precipitation is a scalable technique proportional to the process volume, regardless of its titer, as opposed to the current chromatographic methods that scale stoichiometrically with the reaction

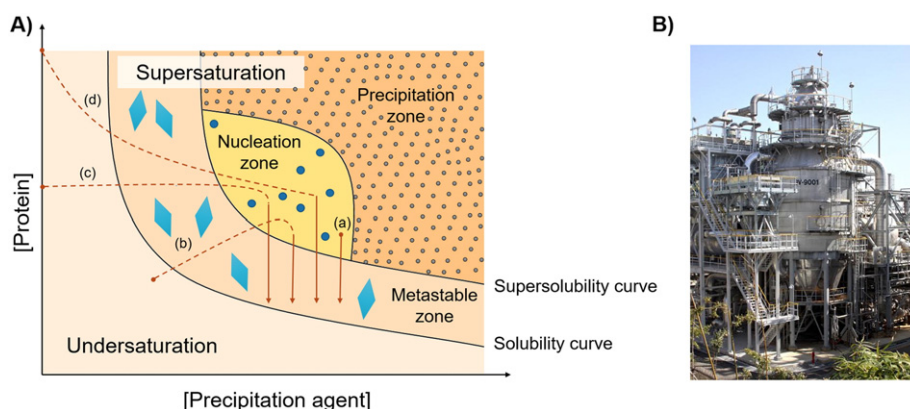


Fig. 1. (A) Qualitative illustration of a protein crystallization phase diagram. The major paths to reach protein crystallization are represented for the four main crystallization methods: (a) microbatch, (b) vapor diffusion, (c) dialysis and (d) free interface diffusion. The solubility curve is defined as the equilibrium between crystals and protein in the solute at a certain concentration. The supersolubility curve separates the condition from which nucleation or precipitation spontaneously occur. Adapted from (Chayen, 2004) (B) Industrial insulin crystallizer from GEA Group Aktiengesellschaft.

between molecules in solution, i.e. titer dependent (Oelmeier et al., 2013). Currently, precipitation can be performed in batch, tubular and continuous stirred tank reactors, giving flexibility to the pharmaceutical industry regarding the working mode (batch or continuous) (Hammerschmidt et al., 2015a, 2015b; Tscheliessnig et al., 2014).

Precipitation conditions can be rationalized according to the interactions predicted to occur between the target and the precipitant. However, in many cases the theory cannot be translated into practice, making this an intrinsic empirical process due to the great amount of interactions prone to occur. Nonetheless, this makes it possible to selectively target the precipitation of a specific protein. On the other hand, in contrast to crystallization, precipitation can cause a protein to lose its activity, therefore it becomes more critical to evaluate the biological activity of the final product.

Introducing crystallization in an industrial environment, requires a crystallizer tank (Fig. 1B), that in the pharmaceutical industry tends to work under batch conditions, and a possible pre-conditioning step (e.g. buffer exchange), as previously reported (Smejkal et al., 2013; Zang et al., 2011). For HTP screening of protein crystallization conditions robotic techniques are used, namely Oryx systems (from Douglas Instruments, Ltd), mosquito systems (from TTP Labtech), Rock Maker systems (from Formulatrix) or Cobra or Phoenix Liquid Handling System (both from Art Robbins Instruments).

Nevertheless, crystallization conditions are established empirically and are not standard for proteins of the same family, which is considered a drawback in industry. Although the same interactions (e.g. hydrophobic, electrostatic) are known to play an important role in protein crystallization by promoting the correct fold (Poor et al., 2015) and preserving the protein structure and function (Baker et al., 2015), the intricacy of all the interactions occurring in the crystallization phenomena is such that the behavior of each protein has to be considered by itself. On the other hand, this singularity can be advantageous when purifying a macromolecule - considering the specificity of the crystallographic conditions for a single protein, it would be beneficial to promote the crystallization of the target protein from the crude extract, leaving contaminants in the supernatant.

3. Established purification processes

Crystallization and precipitation are two unit operation implemented in the purification pipeline of some therapeutic proteins, due to their advantages comparing with traditional DSP unit operations (e.g. chromatography). Crystallization was first employed in insulin purification, while precipitation methods are used for the purification of human serum albumin and erythropoietin.

3.1. Insulin

Insulin is a peptide hormone produced in the pancreas by β -cells as an inactive hexamer, active only when in its monomer forms (5.8 kDa). This hormone is responsible for the uptake of glucose from the blood stream, regulating the carbohydrate and lipid metabolisms, and preventing the liver from producing glucose. Insulin is administered to patients with *diabetes mellitus*, a metabolic condition that leads to high blood sugar levels from long fasting periods in people affected by this condition. Per year 10 tons of insulin are produced, representing a global market of \$5 billion (Kelley, 2007).

In the 1920s the source of insulin was animal pancreas, the first patient being injected with bovine insulin. The first 3-dimensional structure of insulin was unveiled in 1969 by X-ray crystallography in Dorothy Hodgkin's laboratory (PDB file: 1INS) (Adams et al., 1969). As shown by Scott (1934), it is possible to crystallize insulin in conjugation with metal ions as zinc (Scott, 1934). With that in mind, Petersen (1947) patented insulin purification using citrate buffer with two zinc ions per unit cell (0.34% zinc) (Petersen and Schlichtkrull, 1964). However, only in 1969 the first patent reporting insulin crystallization from

an animal source was filed by Eli Lilly, describing what turned out to be the 8.2 process, since the major yield (90–95%) was obtained with the addition of 1 M sodium hydroxide to a 0.5 M acetic acid and insulin solution, at pH 8.2. After 15 min, crystallization occurred spontaneously (Jackson, 1973). In 1978, with the development of recombinant DNA technology, Genentech scientists were able to produce human insulin in *Escherichia coli* cells. This led to Humulin, the first biopharmaceutical molecule produced and commercialized by 1982. Since the first insulin crystallization patent at a liter scale, the industry evolved to 50,000 L tanks used nowadays, thus demonstrating the feasibility of integrating crystallization in a biopharmaceutical bioprocess. The insulin purification process outlined in Fig. 2A shows that crystallization plays a crucial role in the different unit operation, since it is employed at three times in the DSP train. Combining this purification method with recombinant DNA technology manufacturers are nowadays able to commercialize the purest insulin formulation, with purity higher than 98% (Borowicz et al., 2006).

3.2. Human serum albumin

Human serum albumin (HSA) is present in human plasma as a monomer (66 kDa), being the most abundant protein in human plasma at a concentration of 30–50 g/L, and has a serum half-life of approximately 20 days (More and Bulmer, 2013). Being the most abundant protein, HSA represents the determining factor in maintaining the osmotic pressure exerted by proteins in a blood vessel's plasma (oncotic pressure). In addition, due to its ligand binding capacity, HSA has not only the ability to transport (e.g. hormones or fatty acids), but also to affect the pharmacokinetics of some drugs increasing the circulation time (Fanali et al., 2012). HSA is also used as a biomarker for different diseases that include cancer, rheumatoid arthritis, ischemia and severe acute graft-versus-host disease. As therapeutic agent it is used in treating hypovolemia, shock, burns, surgical blood loss, trauma, hemorrhage, acute liver failure, cardiopulmonary bypass, hemodialysis, chronic liver disease, nutrition support, resuscitation, and hypoalbuminemia acute respiratory distress syndrome (Fanali et al., 2012). Per year 500 tons of HSA are produced, representing a global market of \$1.5 billion (Kelley, 2007).

In 1940 HSA was first purified by the Cohn fractionation process, for intravenous use as blood substitute. In this five step ethanol fractionation process, HSA was recovered in the final fraction with 40% ethanol at pH 4.8 (Cohn, 1941). Since then, variations of the Cohn process increased yield and purity. Due to the great interest in HSA as a therapeutic agent, many patents regarding its purification have been filed. In 1979, a patent assigned to Plasmescio AG, reported a purification process inspired in the Cohn method using a heat ethanol/PEG process (Schneider et al., 1979). In this process a 96% recovery yield and <99% protein purity were achieved using 9% (v/v) ethanol, pH 6.5 at 68 °C. More recently, in 2014 a patent assigned to GE Healthcare Bio-Sciences for plasma proteins fractionation with polyacrylic acid precipitation was issued. In this purification scheme, polyacrylic acid precipitation is used as a capture step, prior to the chromatographic polishing step, as it is represented in Fig. 2B. In this case HSA did not precipitate; it was maintained in the supernatant, with a recovery yield of 88%. Next, a chromatography step, affinity or ion exchange, was performed (Alstine et al., 2008).

3.3. Erythropoietin

Erythropoietin, or EPO, is a highly glycosylated protein, it has a molecular weight of 34 kDa and is responsible for erythropoiesis in the bone marrow. In the blood, EPO has a half-life of 5 h (Broxmeyer, 2013; Cheng et al., 2015). As a therapeutic agent, it is most commonly used for anemia treatment. EPO is produced by recombinant DNA (rDNA) technology, making different pharmaceutical agents with different glycosylation patterns that have been approved and are



EPO was first purified by Miyake and co-workers in 1977, in a seven-step purification process, that included ethanol precipitation (Miyake et al., 1977). The overall process consisted of (i) anion exchange

chromatography, (ii) phenol treatment, (iii) ethanol precipitation, (iv) anion exchange chromatography, (v) cation exchange chromatography, (vi) gel filtration and (vii) adsorption chromatography, yielding a final recovery of 21% with 95% purity (Miyake et al., 1977). This study enabled further characterization studies, disclosing EPO amino acid sequence and coding sequence. In 1985, Lin and co-workers were able to isolate the gene encoding for EPO from a genomic phage display library, and produced it in Chinese hamster ovary cells (Lin et al.,

1985). Later in 1987, Amgen conducted a clinical trial using recombinant human EPO in 25 anemic patients with end-stage renal disease (Eschbach et al., 1987). With the trade name Epogen, it was approved by the FDA in 1989 and by EMA in 2007.

The first patent in EPO purification was assigned to Genetics Institute in 1987, with an initial purification step by ethanol precipitation, followed by different chromatographic methods (Hewick and Seehe, 1987). By 2001, a patent assigned to Sterrenbeld Biotechnologie North, regards the differential precipitation of contaminant proteins, leaving EPO in the supernatant, with 100% recovery yield, followed by a series of chromatographic methods for further purification (Carcagno et al., 2001).

4. Purification processes under development

Crystallization and precipitation technologies can overcome some bottlenecks of current DSP unit operations, through their capability to handle high process volume and protein titers. Several efforts are being made to evaluate their potential in replacing standard purification techniques. Currently, the use of crystallization for Apo2L purification is under development. At the same time, the development of both crystallization and precipitation techniques for mAbs purification is being intensively studied.

4.1. Apo2L

Apo2 ligand or tumor necrosis factor (TNF)-related apoptosis-inducing ligand (Apo2L/TRAIL), belongs to the TNF superfamily. The human full length of this transmembrane protein is composed of 281 amino acids and folds as a type II transmembrane protein, being a single-span transmembrane protein with the C-terminal exposed to the extracellular space (Mongkolsapaya et al., 1999; Wiley et al., 1995). The recombinant human soluble Apo2L/TRAIL, consisting of the TNF-homologous portion of the extracellular domain of the full length Apo2L/TRAIL protein, is a 168 amino acid polypeptide with molecular weight 19.6 kDa (Ashkenazi et al., 1999). Apo2L has a potential apoptotic function, identified in various cancer cell lines, but no apoptotic function was yet associated in non-cancer cells (Ashkenazi et al., 1999; Wiley et al., 1995). Therefore, Apo2L has the potential to act as a tumor repressor in tumor patients.

The first Apo2L structure was disclosed in 1999 by X-ray crystallography (PDB file: 1D4V, (Mongkolsapaya et al., 1999)). Small scale crystals were produced using 25% ethylene glycol with 0.1% n-octyl- β -glucoside by vapor diffusion (Mongkolsapaya et al., 1999). With great potential to be used as a tumor repressing agent, and with the capability to be crystallized, in 2010 Genentech patented a new purification process for Apo2L. Crystallization was performed after a cation exchange chromatography. The eluted fraction of Apo2L, in low ionic strength solution (100 mM – 200 mM NaCl or Na₂SO₄) was cooled down to 4 °C enabling spontaneous crystallization. The crystallization occurs with or without anti-solvents (PEG, ethanol, isopropanol, dioxane or methyl pentanediol). In both conditions, the protein purity reported is at least 99% with 95% recovery yield (Flores et al., 2010).

4.2. mAbs

Due to their antibody-antigen interaction, mAbs are broadly employed as therapeutic agents, as diagnostic tools for *in vivo* and *in vitro* analysis, as well as in affinity purification processes (Evans and Syed, 2014; Roque et al., 2007). As therapeutic agents, mAbs are mostly used in oncology and immune disorders, however they are also prescribed for infectious, neovascular, haemostasis and host-to-graft diseases (Elvin et al., 2013). Although different mAbs present a high degree of homology, the conserved domains of mAbs led many companies to define a general purification process based on a common sequence of unit operations, where the protein A chromatography capture step is central (Fig. 3A) (Liu et al., 2010). Even though stable,

reliable, and reproducible, this methodology is considered to be a productivity bottleneck and especially expensive. The bottlenecks associated with protein A chromatography regard low elution pH (2–3) which may lead to ligand leaching and mAb precipitation. Since this is a chromatographic method, it is not able to deal with high process titers (e.g. 10 g/L) or high volumes. In addition the chromatographic process is performed under low flow-rates. On the other hand, this is a stable, reliable, and reproducible method that despite its disadvantages, continues to be preferred over other commercially available options due to the high purification achieved, great versatility and process standardization.

Even though currently mAbs crystallization is not yet a common and implemented procedure, the hope is that this approach can overcome some of the chromatography bottlenecks (Fig. 3B and C). In order to create a rational approach, several studies try to enlighten mAbs behavior under crystallization conditions. In Table 1 a summary of reports in the literature regarding mAbs purification through crystallization and precipitation is presented. A systematic analysis of data available for mAb crystallization conditions, led to the conclusion that PEG is a common precipitating agent for crystallization at room temperature (Table 1) (Baker et al., 2015; Hildebrandt et al., 2015; Huettmann et al., 2014; Oelmeier et al., 2013; Zang et al., 2011). The work of Zang and co-workers (Zang et al., 2011), first demonstrated the feasibility of having a crystallization step in the downstream process of mAbs at a μ l scale. In this study, IgG4 crude extract was purified by protein A chromatography and dialyzed for buffer exchange with 20 mM Tris buffer pH 7.0 before the crystallization step. As precipitant solution, 5% (w/v) PEG 8000, 0.2 M Ca(OAc)₂, 0.1 M imidazole at pH 7.0 was used and crystals were obtained after 5 days at room temperature. After mAb IgG4 solubilization, the final solution showed 90% of protein purity and 31.3% recovery yield. Following this study, Smejkal and co-workers in 2013 (Smejkal et al., 2013) were able to crystallize an IgG1 in 1 h using 5 mM histidine/acetate buffer, 10 mM NaCl, Tris-base up to pH 6.8 at 4 °C. Here, a scale-up from μ l to L in batch conditions was successfully accomplished, with a final protein purity of 96.9% and 90% recovery yield.

Although precipitation is not a conventional purification step in mAbs DSP, precipitation methods are regarded as alternatives. Recent studies show the feasibility of using precipitation as a mAb purification method, resulting in a highly pure and concentrated active protein (Hammerschmidt et al., 2015a, b; Sommer et al., 2015; Tscheliessnig et al., 2014). For example, CaCl₂/cold ethanol precipitation has attracted attention as it can be performed in a simple equipment as stirred tanks and centrifuges. A study from Jungbauer group, demonstrated the possibility of mAb precipitation from mammalian cell supernatant with 78% recovery yield and >99% protein purity (Tscheliessnig et al., 2014). A more recent study, improved CaCl₂/cold ethanol precipitation in a continuous tubular reactor achieving 92% recovery yield (Hammerschmidt et al., 2015a). As for crystallization, PEG-based precipitation systems are frequently also tested with mAbs. Application of high-throughput screening to a rapid identification and selection of PEG precipitation conditions using only μ l volumes of protein is usually the initial approach in these studies, since it enables a rapid selection from a wide range of different conditions (Knevelman et al., 2010). A comparison between batch and continuous PEG precipitation, demonstrated that both methods lead to >90% recovery yield and >96% protein purity without changes in mAbs secondary structure (Hammerschmidt et al., 2015b). Combination of CaCl₂/PEG precipitation for direct mAb purification from the cell culture supernatant can also be used. The CaCl₂ precipitation promotes the precipitation of high molecular weight contaminants, while PEG precipitation was used for mAb capture (Hammerschmidt et al., 2015a). This method allowed 80–96% recovery yield. Additionally, the use of linear or branched PEG in mAb precipitation was evaluated, showing the feasibility of using both for purification purposes with recovery yield >95% (Sim et al., 2012b).

Other concept in purification is affinity precipitation. In this system a protein or peptide with affinity towards the mAb is placed in the cell

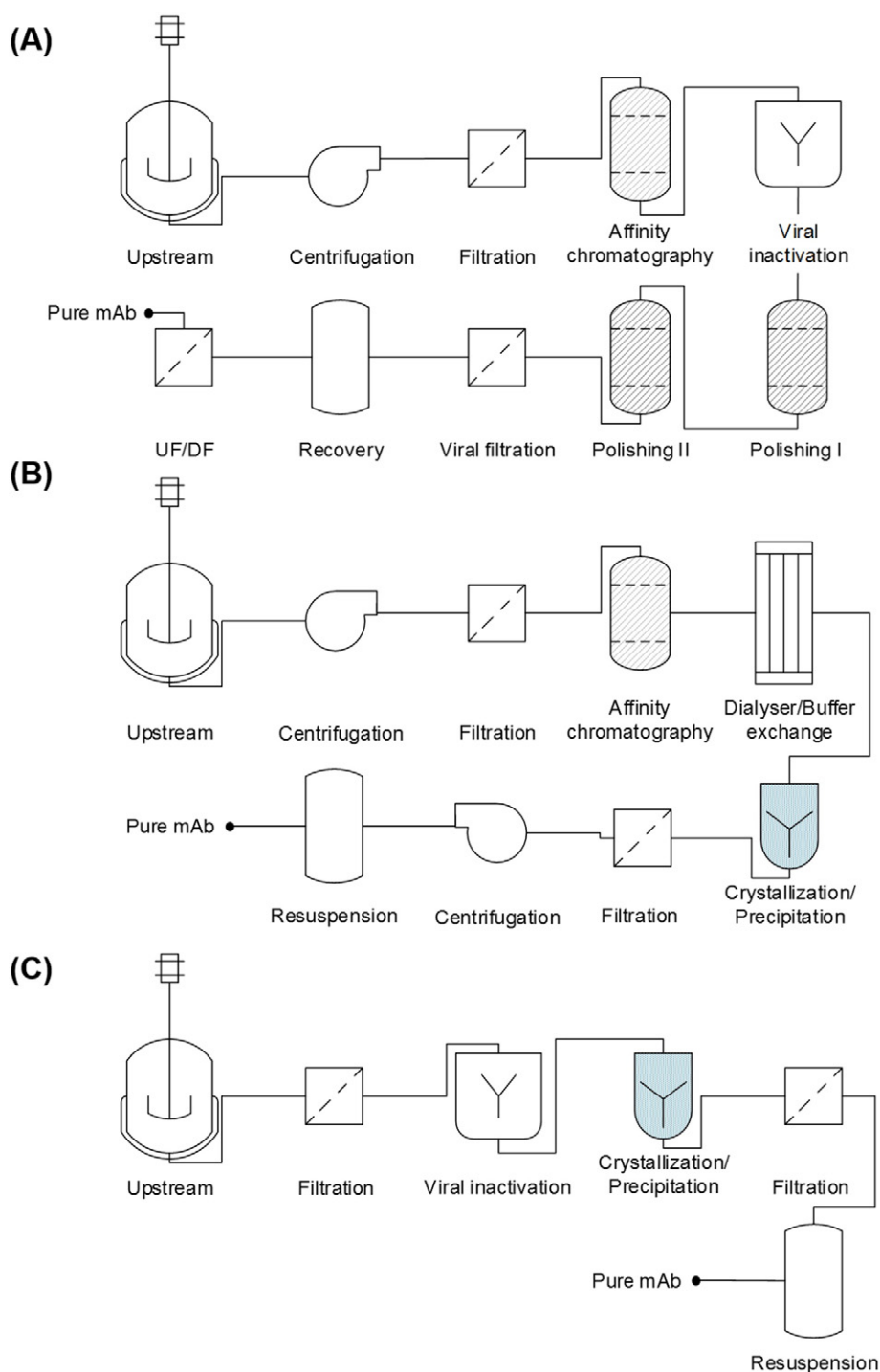


Fig. 3. Schematic representation of different purification processes for mAbs using chromatographic supports or crystallization/precipitation techniques. (A) Standard mAbs purification by chromatographic methods. (B) mAbs purification using crystallization/precipitation techniques (Smejkal et al., 2013; Zang et al., 2011). (C) A proposed purification process for mAbs relying on crystallization/precipitation techniques.

culture supernatant and the precipitation of the complex is promoted (Handlogten et al., 2014; Janoschek et al., 2014; Sheth et al., 2014a, b). In addition, other precipitants had been used in combination with chromatographic methods, as polishing steps for impurity removal from the elution fraction of protein A chromatography (Chollangi et al., 2015).

5. Concluding remarks and future trends

The biopharmaceutical sector has the highest estimated growth rate for the next five years. Given this scenario, the biopharmaceutical

industry will be faced with a higher production demand in the near future. Developments in the USP have already been successfully accomplished, shifting the bottleneck towards DSP. Although progresses in better performance of cell culture and growth conditions should continue to occur, DSP must explore alternative methods. Crystallization and precipitation have the great advantage of scaling-up with the process volume, instead of the process titers as in chromatography processes. In Table 2 a comparison between protein A chromatography and crystallization/precipitation in terms of scaling-up and purification criteria are presented.

Table 1

Summarized review of purification processes under development using crystallization or precipitation as unit operations in mAbs DSP. The mAb class and source where it was produced and any pre-treatment used prior to the crystallization or precipitation are discriminated. The protein purity and purification yield are described according to the crystallization or precipitation condition used for purification. ND: Not determined.

MAb class	Source	Purity ^a	Yield ^b	Crystallization/Precipitation conditions	Reference
IgG 4	CHO cell supernatant, dialyzed	90%	31%	5% (w/v) PEG 8000, 0.2 M Ca(OAc) ₂ , 0.1 M imidazole, pH 7.0	(Zang et al., 2011)
IgG 1	CHO cell supernatant, dialyzed	93% 97%	88% 90%	5 mM histidine/acetate buffer, 10 mM NaCl, pH 6.8	(Smejkal et al., 2013)
IgG 1	CHO cell supernatant, protein A purified	90%	96%	250 mM CaCl ₂ , PEG 20000-PEG 2000	(Sommer et al., 2014)
IgG 1	CHO cell supernatant, partially purified	ND	>93%	50 mM citrate buffer, pH 6.0 with 15% (w/w) PEG 4000	(Oelmeier et al., 2013)
IgG 1	CHO cell supernatant, protein A purified	ND	92%	250 mM CaCl ₂ , 25% (v/v) Ethanol	(Hammerschmidt et al., 2015b)
IgG 1		99%	84%	Caprylic acid precipitation: 1% caprylic acid in 100 mM sodium citrate pH 4.5 PEG precipitation: 14% (w/v) PEG 6000 pH 7.0	
	CHO cell supernatant	>95%	69%	Caprylic acid precipitation: 1% caprylic acid in 100 mM sodium citrate pH 4.5 PEG precipitation: 14% (w/v) PEG 6000 pH 7.0	(Sommer et al., 2015)
IgG 1				CaCl ₂ precipitation: 120–150 mM CaCl ₂ , 300 mM Na ₂ HPO ₄ , 4 mM PO ₄ ³⁻ pH 8.5 Cold ethanol precipitation: 25% (v/v) ethanol	
		>95%	59%	CaCl ₂ precipitation: 120–150 mM CaCl ₂ , 300 mM Na ₂ HPO ₄ , 4 mM PO ₄ ³⁻ pH 8.5 Cold ethanol precipitation: 25% (v/v) ethanol CaCl ₂ precipitation: 120–150 mM CaCl ₂ , 300 mM Na ₂ HPO ₄ , 4 mM PO ₄ ³⁻ pH 8.5 Cold ethanol precipitation: 25% (v/v) ethanol	
IgG 1	CHO cell supernatant, protein A purified	70–90%	> 95%	9–10% (w/v) Branched PEG 600 or Linear PEG 4000/PEG 6000	(Sim et al., 2012b)

^a Ratio between the target protein concentration and protein concentration.

^b Percentage of purified protein in terms of initial amount of protein.

The implementation of protein crystallization or precipitation in purification processes will include, the high-throughput screening of different precipitants, polymers and salts, at different conditions (e.g. pH and temperature), followed by a scaling-up of the optimized methods. The great innovation in this field of purification may arise from the integration of different and new additives enhancing the crystallization efficiency and decreasing the process time (Kowacz et al., 2015; Ribeiro et al., 2014) or the integration of affinity molecules to target the biopharmaceutical and promote its selective precipitation (Janoschek et al., 2014; Sheth et al., 2014b).

Crystallization and precipitation are two unit operations already implemented in the downstream pipeline of therapeutic proteins. Indeed, crystallization is successfully used as a final polishing step, enabling at the same time greater protein purity and concentration (Jackson, 1973). On the other hand, precipitation can also be performed as an initial capture step, either precipitating the target protein or contaminants, or as a final purification and concentration step (Alstine et al., 2008; Carcagno et al., 2001).

In order to be considered as alternatives to the current and well implemented chromatographic steps in therapeutic protein purification, a method has to be able to handle high volume and protein titer, be scalable, and economically viable at an industrial scale. In this regard, both techniques are able to cope with the biopharmaceutical industry demands.

Table 2

Comparison between protein A chromatography and crystallization/precipitation unit operation in mAbs DSP regarding scaling-up and purification criteria.

	Protein A chromatography	Crystallization/Precipitation
Capacity	40–60 g IgG/L resin	Up to 150 g IgG/L precipitant
Scale-up	Process volume	Stoichiometrically
Cost	5000–14,000 €/L resin	15–140 €/L precipitant
Yield	>95%	>90%
Purity	>95%	>95%
Industrial equipment	Chromatographic column	Stirred tank

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